

FieldSense TT-120 Temperature Transmitter -- Technical Manual

PRODUCT OVERVIEW

The FieldSense TT-120 is a state-of-the-art temperature transmitter designed specifically for the rigorous demands of wastewater treatment applications. Ideal for monitoring temperatures in biological treatment processes such as activated sludge treatment, the TT-120 also provides critical temperature data for pump motor overheating protection, enhancing predictive maintenance strategies. The robust design and advanced sensing technology ensure reliable performance in harsh environments, making it an essential instrument for maintaining optimal operational conditions.

TECHNICAL SPECIFICATIONS

- Measurement Range: -40°C to 120°C (-40°F to 248°F)
- Accuracy: $\pm 0.5^{\circ}\text{C}$ ($\pm 0.9^{\circ}\text{F}$) within the range of -10°C to 100°C; $\pm 1^{\circ}\text{C}$ ($\pm 1.8^{\circ}\text{F}$) outside this range
- Sensor Type: PT100 Class A RTD (Resistance Temperature Detector)
- Process Connection: 1/2" NPT or flanged (2" and 4" options available)
- Output Signal: 4-20 mA, HART protocol compatible for digital communication

CALIBRATION PROCEDURE

1. Preparation:

- Ensure the FieldSense TT-120 is powered off and disconnected from any power sources.
- Gather required tools: calibration thermometer (with accuracy of $\pm 0.1^{\circ}\text{C}$), multimeter, and calibration software (if using HART features).

2. Connection:

- Connect the calibration thermometer to a stable temperature source, such as a water bath or temperature calibrator.
- Connect the TT-120 to the multimeter and ensure all connections are secure.

3. Calibration Steps:

- Power on the TT-120 and allow it to stabilize for approximately 15 minutes.
- Immerse the TT-120 sensor and the calibration thermometer in the temperature source.
- Monitor the readings from both devices until they stabilize.
- Adjust the TT-120 output signal using the calibration software or onboard adjustments until the output matches the calibration thermometer reading.
- Repeat this process at three points: low (-10°C), mid-point (50°C), and high (100°C) for comprehensive calibration.
- Once calibrated, document the settings and any adjustments made.

4. Finalization:

- Disconnect all equipment and re-install the TT-120 in its operational environment.
- Power on the device and verify functionality before returning it to service.

ERROR CODES

- Error Code: E01 - SENSOR FAILURE

Cause: The PT100 sensor is not providing a valid reading.

Resolution: Inspect the sensor connections and replace the PT100 sensor if necessary.

- Error Code: E02 - OUT OF RANGE

Cause: The measured temperature exceeds the specified limits of the TT-120.

Resolution: Verify the process temperature and ensure it is within the operating range. Check for potential thermal shock or equipment malfunction.

- Error Code: E03 - COMMUNICATION ERROR

Cause: Failure in HART communication with the connected devices.

Resolution: Check wiring and connections. Ensure proper setup of the HART communication settings. Restart the instrument if the issue persists.

- Error Code: E04 - POWER SUPPLY FAILURE

Cause: Insufficient voltage detected at the power input.

Resolution: Inspect power supply connections and verify voltage levels. Replace power supply if faulty.

MAINTENANCE

Routine Checks:

- Monthly: Inspect the physical condition of the FieldSense TT-120 for signs of wear, corrosion, or damage.
- Quarterly: Perform a visual inspection of the wiring and connections for signs of fraying or loosening. Ensure that all external connections are secure.
- Biannually: Conduct a functional test using a calibrated thermometer to verify accuracy. Document any discrepancies.

Troubleshooting:

- If readings are erratic or inconsistent, check for environmental influences such as nearby heat sources or airflow disruptions.
- Clean the sensor area to remove any deposits or debris that may affect readings.
- If the unit continues to show errors or malfunction, consider consulting FieldSense technical support for further diagnostics or service options.

The FieldSense TT-120 is engineered for durability and precision, ensuring it remains a cornerstone of effective wastewater treatment and predictive maintenance strategies. Regular maintenance and calibration will help maintain optimal performance and longevity of the device.